Wednesday Oct-07-2026

Flowserve Facility:

Flowserve Cookeville Operations
1978 Foreman Drive
Cookeville, TN 38501

cmtrs@flowserve.com

To:

Valve Summary

SERIAL #	HEAT #	PO #	Order #	Description	Valve Type	Class	End Connection
61-677610	1637Z	1158721	1109134	4" 2345		900	FLG

FORM QF-MT VER 4

FLOW LINK SYSTEMS PRIVATE LIMITED
FOUNDRY UNIT - II
65 / 2-A,B & C,PUDUPALAYAM B.P.O
AVINASHI Tk, COIMBATORE - 641 654
INDIA

INSPECTION CERTIFICATE

(EN 10204:2004-3.1)

TC NO : **X-1866**
DATE : **25-Jul-2022**

CUSTOMER
FLOWSERVE INDIA CONTROLS PVT LTD,
B8, MMDA Industrial Area,
Maraimalai Nagar-603209
Kancheepuram, INDIA

SPECIFICATION REFERENCE
1. ASTM A216-21 Gr.WCC
2. Customer Spec. RMC 90012 Rev. 14

HEAT NO.
1637Z
Date of Pouring
08-Jun-2022
FOUNDRY MARK
FLS

PO NUMBER & DATE	ORDER ACCEPTANCE NO
10104112 Dt. 21.03.2022	22-1087

CHEMICAL COMPOSITION % wt

MELTING PRACTICE : **Steel melted using Electric Induction furnace, and fully killed using Aluminium.**

SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.200	0.365	0.008	0.015	0.891	0.065	0.097	0.020	0.003	0.015	0.045	--	0.20	0.38

Heat Treatment : NORMALIZING: Temperature raised to 920°C, Soaked for 4 hours and then atmospheric air-cooled

Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)

MECHANICAL PROPERTIES

TEST TYPE	Tensile as per ASTM A370-21 / ISO 6892-1-2020						Impact (CVN) ISO 148-2016/ASTM A370-21		BEND TEST		HARDNESS
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d	% Elongation 5d	% RA	Test Temperature (at °C) J Values	Test Temperature (at °C) J Avg.	ANGLE	D	(HBW / HRBW)
Minimum	40	--	70	22	--	35	--	--	--	--	--
Maximum	--	--	95	--	--	--	--	--	--	--	--
Achieved	47.0	--	76.4	35.8	--	60.7	--	--	--	--	150

DETAIL OF CASTINGS

S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV	PART NO	POURED QTY	DESP. QTY
1	01	4" 900 FE BODY	D-487876 / 20	N2757928-60	7	7

MTC - REVIEWED & ACCEPTED
10139322 FICPL QC

REMARKS:
1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block.
2. *RE = Ni + Cr + Mo +V +Cu
3 ** $CE = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$
4. The above stated KSI values have been calculated based on original tested N/mm² values.

CERTIFIED THAT
1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO.
2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1.
3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.

PREPARED BY:
S.KARTHICK

CHECKED BY:
A.LOGANATHAN
LAB CHEMIST

M.S.DHANABALAN
ASSISTANT MANAGER - QUALITY
MANUFACTURER'S AUTHORIZED INSPECTOR